

PREDICTING EMPLOYEE ATTRITION USING MACHINE LEARNING

Machine Learning Model Comparison for HR Decision Support

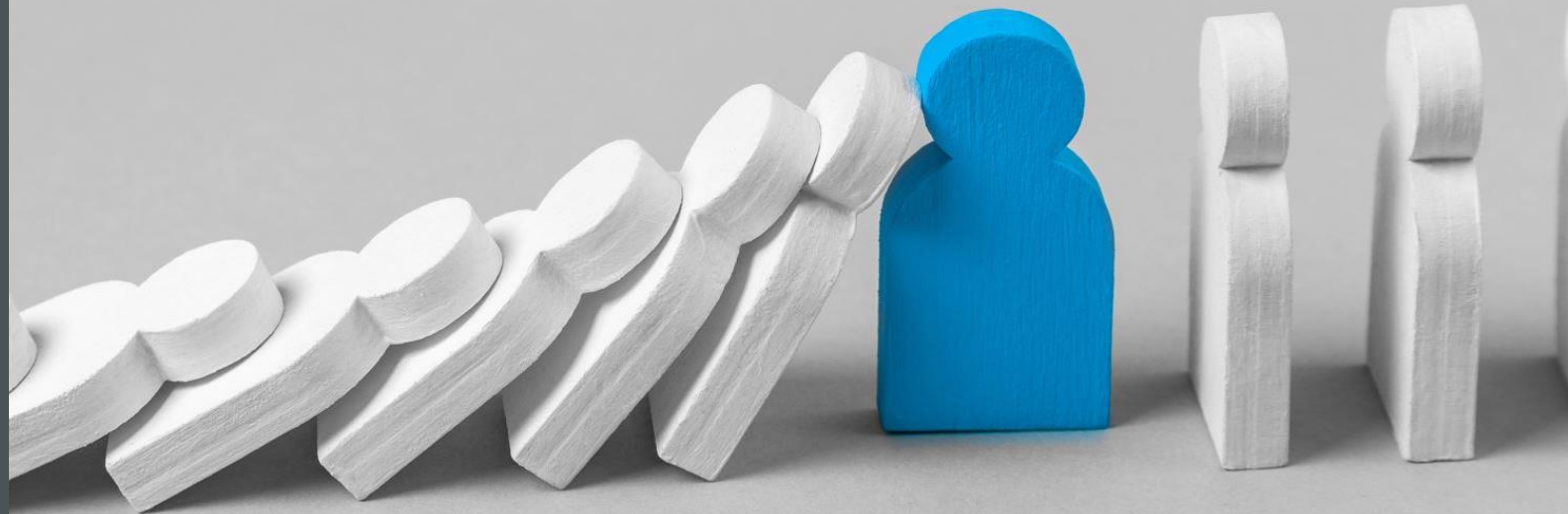
Loren Martin

Python • Random Forest • AdaBoost •
Power BI

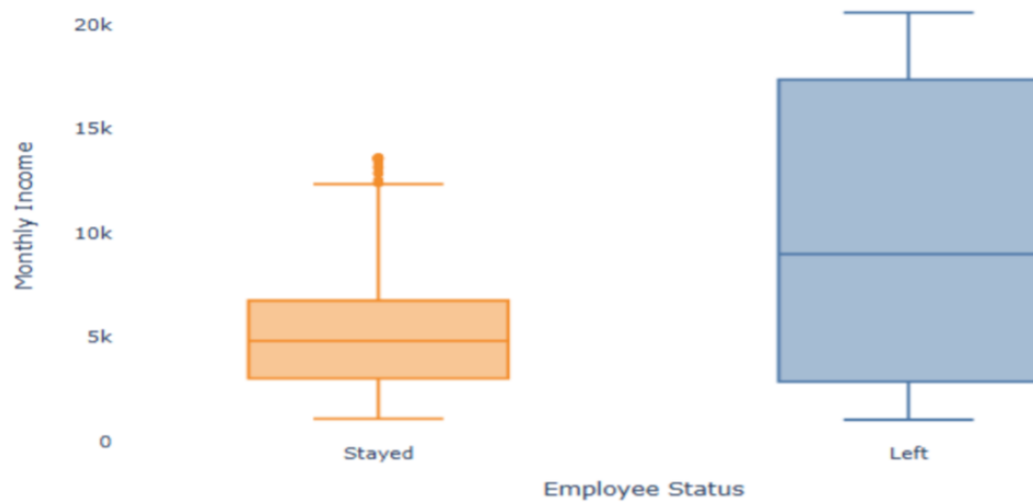


BUSINESS PROBLEM AND OBJECTIVE

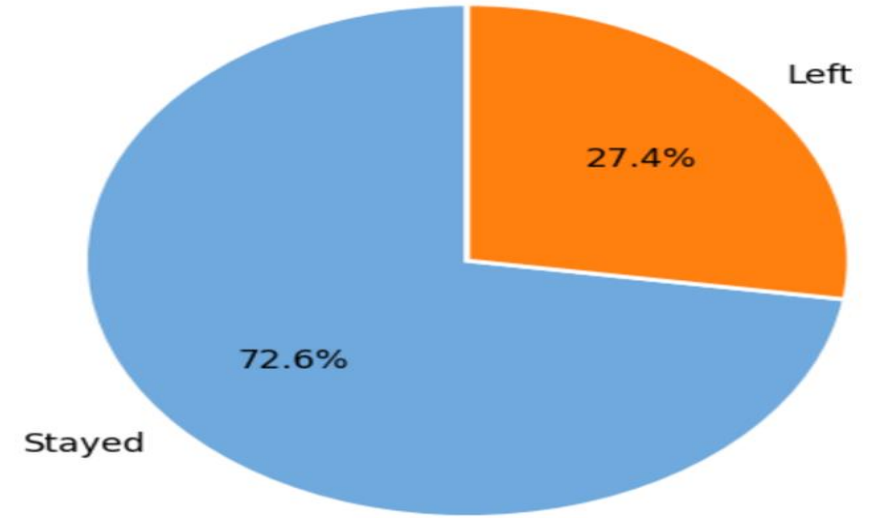
- **Business Problem:** Employee attrition creates costs associated with recruiting, training, and lost productivity.
- HR leadership needs a way to identify employees who may be at greater risk of leaving.
- **Objective:** Determine whether workforce characteristics can predict employee attrition.
- Random Forest and AdaBoost were compared to determine how effectively each model identifies employees at risk.



Monthly Income by Employee Attrition



Employee Attrition Rate



EXPLORATORY DATA ANALYSIS AND FEATURE ENGINEERING

- **27.4% of employees left the organization, highlighting a meaningful retention concern.**
- **Monthly income showed variation between employees who stayed and those who left, although substantial overlap suggests it should be considered alongside other workforce characteristics.**
- **Multiple workforce characteristics were evaluated and prepared for use in predicting employee attrition.**

PREDICTIVE MODEL PERFORMANCE

- **Both models achieved approximately 85% testing accuracy**, demonstrating useful predictive capability for identifying attrition risk.
- **Random Forest identified more employees who ultimately left** (59% recall), making it more useful when HR's priority is reaching more at-risk employees.
- **AdaBoost produced more reliable positive predictions** (92% precision), helping HR focus retention resources on employees most likely to leave.
- **The best model depends on the business priority:** broader identification of attrition risk or more targeted intervention.

Employee Attrition Prediction Dashboard





Main conclusion

Historical workforce data can help identify employees at greater risk of attrition with useful predictive accuracy.



Business opportunity

Predictive models can help HR identify employees who may benefit from earlier retention interventions.



Random Forest recommendation

Prioritize Random Forest when HR's goal is to identify a broader share of employees at risk of leaving.



AdaBoost recommendation

Prioritize AdaBoost when HR's goal is reducing false positives and targeting retention efforts more selectively.



Recommended use

Use model predictions as a decision-support tool alongside workforce characteristics to guide proactive retention planning.



Decision Consideration

Model selection should reflect HR's retention priorities and available resources.

CONCLUSIONS & RECOMMENDATIONS

REFERENCES

Scikit-learn developers. (n.d.). Precision-recall. Scikit-learn.
https://scikit-learn.org/stable/auto_examples/model_selection/plot_precision_recall.html

Southern New Hampshire University. (n.d.). HRData1.csv [Data set]. DAT 430 Project Two Virtual Lab.

Southern New Hampshire University. (n.d.). HRData2.csv [Data set]. DAT 430 Project Two Virtual Lab.

Ruehl, E. (2024, April 15). Attrition: Definition, types, causes & mitigation tips. SHRM. <https://www.shrm.org/topics-tools/news/employee-relations/attrition-definition-types-causes-mitigation-tips>